

SUNY Polytechnic Institute

Computer Science 498

Dr. R. Sarner

Capstone CS 498/IS 495

I. Description

The capstone project is an opportunity for the student to demonstrate the breadth and depth of what has been accomplished during the undergraduate experience. It consists of: (1) defining an appropriate project, (2) carrying out that project, (3) at the conclusion giving a presentation describing the findings and demonstrating deliverables, and (4) submission of a reflective paper as described below.

II. Project Definition (Proposal)

Prior to granting permission to enter CS 498 or IS 495 students are required to submit a one page proposal describing the project and the deliverables. I review the proposal with the following considerations in mind: (1) is there space remaining in my capstone sections for the target semester, (2) is the project domain something that I have enough familiarity with to be able to appropriately evaluate, (3) is the scope of the project reasonable - neither too ambitious nor too trivial. If I am satisfied that the proposal meets all these criteria I approve it and ask the department's administrative assistant to enter the instructor's permission into Banner.

III. Semester Progress Check-In

During the semester students are asked to check in with me monthly. This is to ensure that the project remains on target and is likely to be completed in a timely fashion. These meetings can be in person, by phone, or by Zoom; they are typically quite brief. Just the fact that the student needs to come in and brief me tends to reduce the tendency to procrastinate.

IV. Presentation

The presentation at the end of the semester typically occurs during the last week of classes or during finals week. A mutually convenient time is scheduled and the presentation can be either in person or via Zoom. If the project produced some working item - typically some software, application or some hardware, then the presentation should describe the project and demonstrate how it functions. If the project was more theoretical or research-based, then the results of that endeavor should be presented. In the course of the presentation the student should describe problems that were encountered, how they were overcome, what was learned from the experience, and what would be done differently if the project was redone.

V. Paper

The paper that is turned in is a "reflective paper". It should cover the following items:

- (1) Description of the project
- (2) Literature review (if appropriate) - what has been done in this area previously and with what results?
- (3) Description of the development process - describe what was attempted, what problems were

encountered, how they were resolved.

(4) If hardware/software was developed, description of the deliverables including screen shots or pictures that demonstrate how it functions. If a research-based project, describe the findings and their significance.

(5) Reflective conclusion - what did you learn from this project. What would you do differently if you were to start this over? In what ways could this project be further enhanced?

(6) If software - provide all code either displayed in an appendix or provided electronically.